

FROM PARTITIONS TO RANKS, FROM MODULAR FORMS TO HARMONIC WEAK MAASS FORMS AND BEYOND

CLAUDIA ALFES (BIELEFELD)

The partition function, which counts the ways to write a natural number n as a sum of positive integers, is a classical meeting point of combinatorics and the theory of modular forms. The generating function of the partition function is essentially the reciprocal of Dedekind's eta function. Its modularity underlies the Hardy-Ramanujan asymptotic formula, Rademacher's exact formula, and Ramanujan's congruences for the partition function.

When seeking a combinatorial explanation for Ramanujan's congruences modulo 5 and 7, Dyson introduced the rank of a partition: It is given by the largest part minus its number of parts. The associated generating series turns out to be not a classical modular form, but a harmonic weak Maass forms (as discovered by Bringmann and Ono).

In my talk I will trace this development from the partition function to the rank function, and from modular forms to harmonic weak Maass forms before turning to recent joint work with Ken Ono and Ashvin Swaminathan on higher Dyson ranks. These natural generalizations are not expected to fit the classical harmonic weak Maass framework. Nevertheless, functions built from their root-of-unity specializations possess hidden elliptic symmetries. This provides a new continuation of the classical story.

I will also discuss the role of human-AI collaboration and the formal verification of key identities in Lean.